

7.3 Tasks' Definition

REMINDER: It shall be taken into consideration that all tasks in the Mars Yard will be performed separately (i.e. Science Task with Collection, Navigation, Maintenance). Each team will be assigned a time slot for execution of each task. Changes of rover's configuration are allowed (between the tasks).

7.3.1 Science Task

Science Task consists of the following sub-tasks: * Scientific Exploration * Surface Sampling * Deep Sampling * Astro-Bio

Descriptions of these sub-tasks are provided in this section.

7.3.1.1 Scientific Exploration Sub-Task

The aim of the Science Task is to demonstrate the understanding of performing scientific exploration in planetary geology conditions. The workflow of the Scientific Exploration Sub-Task is as follows: 1. Analysis of previous relevant work about the Martian region to be explored (literature review) 2. Careful analysis of provided data about the Mars Yard 3. Preparing a scientific exploration plan that considers the requirements of the data collection 4. Then, execution of the science exploration plan on the Mars Yard during the ERC field trials (Science Exploration Sub-Task) 5. Data analysis 6. And finally, delivering a final report

The Scientific Exploration Sub-Task will be divided into three parts. 1. “Science: Regional Geology Analysis” is graded as a part of the “Final Report” that is part of the ERC finals 2. “Science Planning” that is required to be delivered about a week before the ERC finals 3. “Scientific Exploration” is to be submitted up to 2,5 hours after the ride through the Mars Yard

All reports shall be submitted as separate PDF files using the links listed in the section number 6.4. The PDF file should be signed in the following format `{[name of the team]_[report_name]}`. If you do not follow this naming instruction, 5% points will be deducted from your final grade. Please note that the maximal length of the answer to each question will be strictly enforced (material exceeding the allowed number of characters will not be read and assessed). The Task documentation is scored according to Appendix 2 – Scoring. Late submissions will be penalised (for part 1 and 2 every 24 hours means 20% points less, for the part 3 each 60 minutes delay is 20% less – penalties are proportional).

At some point after the finals, every team may request feedback with the Science Team after the ERC (delivered remotely at a mutually convenient time).

Materials provided by the ERC organizer You will be provided with the following data based on which you will prepare your reports: 1. Detailed instruction (below) 2. An approximate location of the Mars Yard on the surface of the Red Planet. Please note that the Mars Yard does not copy any area of the real Mars but is an artificial model that represents the entire region. This information should help you to better understand the geological context of geomorphological forms present on the Mars Yard 3. Drone images

of the Mars Yard along with the digital elevation model of the area that will be delivered at the latest three weeks before the finals of ERC 4. A non-compulsory workshop that will introduce various topics helpful to accomplish Science Tasks (link: <https://www.youtube.com/live/QNf3-r8kX6g>) 5. Support of our Science Team through the dedicated contact platform.

Required documentation:

Part 1: “Science: Regional Geology Analysis” The goal is to select and shortly describe (in a form of a map and text) a best landing site on a pre-defined region on Mars. Selected landing site must be geologically interesting (within a rover traverse up to 30 km), possible to land there based on currently available techniques (appropriate atmospheric thickness, appropriate landing area) and safe to access by a rover (local slopes inclination $<20^\circ$, lack of numerous large boulders, reasonable thermal conditions for a rover to survive winter). To prepare your analysis you should use Google Earth Pro (choose Mars as a base planet) or, if you are more ambitious, JMars. The report should consist of two parts fitting on two pages A4: 1) A composite map fitting on an A4 format and consisting of a 1) location map showing selected landing site in a global scale, 2) a regional geological map (based on an appropriate map from the Astrogeology Science Center publications) and 3) a close-up map based on CTX or Hirise photos, with a clearly marked a) landing ellipse, b) planned rover path, c) three regions of interest 2) A report: up to 800 words long technical (why is it safe to land there) and scientific (why is it worth landing there – especially in reference to the three regions of interest) justification of the selected area.

The region selected for the ERC2026 is within the deepest impact basin on Mars: Hellas Planitia (± 100 km from -36.409170° 89.679479°). You should expect an interaction between volcanic features (from nearby Hadriacus Mons) and features related to past and contemporary ice presence in the subsurface. Our Mars Yard will be roughly modelled on this geological area on Mars.

Suggested reading: * <https://pubs.usgs.gov/imap/i2694/> * <https://www.sciencedirect.com/science/article/pii/S0169555X24003787?via%3Dihub> * <https://www.sciencedirect.com/science/article/abs/pii/S0032063308002432> * <https://www.sciencedirect.com/science/article/abs/pii/S0019103522004912?via%3Dihub>

Part 2: “Science Planning” The goal is to analyse the Mars Yard as a “landing site” and design a scientific mission in this area. This task’s documentation is based on information (drone footage and digital elevation model) provided no later than 3 weeks before the ERC finals, the report is due 1 week before ERC finals as a pdf file. Science Planning report must contain detailed information on the topics listed below.

1. **Geological Map of the Mars Yard** including a coherent geological interpretation of the ‘landing site’. Outline of all geological features visible on the Mars Yard with a focus on what they are, how they were formed, and what their relative ages are (assigned from the oldest to the youngest). The map should fit on an A4 format of a page. Preparing a good geological map of the area is the most important part of the scientific part of the Science Task: without a good map you will not be able to obtain good grades from other tasks. An example of a geological map: Geologic Map of Jezero Crater and the Nili Planum Region, Mars by V.Z. Sun and K.M. Stack can be found here: <https://pubs.er.usgs.gov/publication/sim3464>

2. An exploration plan: Based on the prepared geologic map, and the understanding of geological development of the area, you should prepare a detailed plan of scientific exploration and properly justify your decisions. This section should be no longer than allocated number of characters (with spaces) and include assigned number and type of figures. Longer descriptions WILL NOT BE READ, more than one (possibly composite) figure will not be taken into account.

- **a)** 1st paragraph: stating a subject to be studied. e.g., Relative ages of Aeolian features present at the landing site. (up to 500 characters with spaces)
- **b)** 2nd paragraph: explaining why we should know that. e.g., determining the geological history of the landing site, and establishing when (and maybe why) dunes became immobilized. (up to 1000 characters with spaces)
- **c)** 3rd paragraph: a falsifiable hypothesis: e.g., an impact crater is younger than the dune located next to it. (up to 500 characters with spaces). Wiki has a good article on what is a good hypothesis: <https://en.wikipedia.org/wiki/Hypothesis> You should particularly make sure your hypothesis is fulfilling requirements of the Scientific Hypothesis as described in this article. Hypothesis should be related to the geologic analysis present in your map, and cannot be ridiculously obvious like: "is this object a rock?", "is this dune made of sand?". Every team may ask once (through a direct message through the selected communication channel e.g., slack) if your hypothesis is too ridiculously obvious)
- **d)** 4th paragraph + one A4 figure (pdf format): predictions that will test the hypothesis; what is expected to be seen in the field, and how it will help the team sustain or overrule the hypothesis e.g., we will go to the slope of the dune facing the crater and examine how material ejected from the crater interacts with the dune. If we find ejected material on the dune's surface, we will know that the crater is younger than the dune → it therefore supports our hypothesis. If we will find that material ejected from the crater is covered by the dune, we will know that the dune is younger than the crater → our hypothesis will be rejected. A figure can be used to illustrate the expected results (up to 1500 characters with spaces)
- **e)** Relevant scientific references that were appropriately cited in the text (do not cite Wikipedia or NASA web pages – we want to see you read papers or book chapters).

Part 3: “Scientific Exploration” The goal is to: 1. Verify the hypothesis stated in the Scientific Plan. 2. Observe the surroundings for potentially scientifically interesting objects that will be placed around the Mars Yard (e.g., atypical rocks, meteorites, pieces of spacecraft, possible USO (Unexpected Standing Objects)).

Figure 2: Example of Scientific Exploration based on ERC 2024. Start from S1 point, exploration of the Mars Yard with discovery of easter eggs and geological features, finally return to S1.

Deadline for submitting the report: up to 2.5 hours after completing the traverse (refer to §6.4 for the delivery link).

Hypothesis verification: Aim: an attempt to verify your hypothesis described in your Scientific Plan Report. We will judge you on the quality of your data analysis against the hypothesis. It is fine to reject the hypothesis, it is also ok to determine that the data was not sufficient to properly test the hypothesis

(although in this case you should explain what else would be needed to test it). Within your explanation you should include three annotated photographs from the rover showing geologic features that you discuss in the report. You should discuss how this new knowledge influences the understanding of the geology of this area and modify the geological map appropriately. You should note the location of your observations on the geological map and update geological map including all new or modified features (there will be some!!). Please make sure that the new/ changed features are clearly marked on the map.

USO: Unexpected Standing Objects (up to 3 objects): during your traverse you should identify and describe “unexpected objects” identified on the Mars Yard. When doing real science (on Earth and on other planetary bodies) usually the most important science is related to finding new, unexpected things while looking for other stuff. This is why it is crucial to pay attention to your surroundings while traversing Mars. Unexpected items can include: a) rocks that are not similar to other rocks present on the Mars Yard – it may be an important ore for future use, b) signs of extra-terrestrial life (you never know – maybe you will find something), c) signs of technological civilization (either human – related or not), and d) observations of any geological activity. You should identify up to 3 objects and prepare: 1) a short description of the object and 2) an ad-hoc hypothesis about the nature of this object, 3) an image of the object, and 4) it should include a map (drawn on the digital elevation map provided to you by the Organiser) with your traverse and the location of all found objects clearly marked on your updated geological map. The description of each object should be no longer than 350 characters (with spaces) plus one annotated photograph, and up to 3 objects will be graded. Do not afraid to be funny in your descriptions – the judges will need some entertainment during grading.

The report should be no longer than 3000 characters (with spaces) and include no more than 3 figures/ annotated photographs with proper descriptions (figure caption included in the character limit) + an updated geological map.

7.3.1.2 Surface Sampling Sub-Task

The main goal of this sub-task is to collect surface samples of the Martian soil that will be sent back to Earth for further analysis. Surface and Deep Sample Collection Sub-Tasks will be performed in the same time slot. The task scenario is as follows: 1. Arrive to the sampling location 2. Take a photo before sampling is started 3. Perform sampling (regolith and rocks samples in case of surface sampling) 4. Place the collected sample/s in the onboard container 5. Measure weight of the collected sample/s 6. Take a photo after sampling is completed 7. Drive to the start/finish line

General requirements for the surface sampling are as follows: 1. The rover shall be equipped with a manipulation device that is able to pick up the surface samples from the Mars Yard and place these in the on-board container 2. The rover shall be equipped with a container/s allowing for the stable transport of the collected material over challenging terrain 3. The container/s shall keep the collected samples in a selected position and orientation, and it shall prevent any movement 4. Regolith and rock samples (both!) must be collected by the rover. Both mass (>100g as a goal) and size (in case of rocks the largest dimension >150mm as a goal) requirements are applicable to assure that scientifically valid samples and enough material are transported back to Earth. 5. The collected samples (qty=2) must be further disposed in the onboard containers and transported to the finish line (Martian Lander Module – MLMOD). 6. Weight of the samples must be measured by the onboard system of the rover. 7. Special

tool or tools must be used to perform the tasks, e.g., grippers assembled on the robotic arm. 8. Weight and dimensions of the collected sample will be checked by judges at the finish line.

SURFACE SAMPLES COLLECTION PART – GENERAL COMMENTS: * Surface and Deep Sampling Sub-Tasks will be performed in a single run and sequence (one after another, sequence to be selected by teams) * Sampling zones will be predefined by judges. * Remember that in the collection part both surface and deep samples shall be collected! * Remember that in the surface collection part both regolith and rock samples shall be collected! * Remember about taking pictures before and after collecting samples * Measure weight of the collected sample/s. * Science Surface and Deep Sample Collection Sub-Tasks are designed to solely check rover's ability to perform collection of various samples. These samples are not part of the Scientific Exploration activities (shall not be included in the report as the sampling location is defined by the organizer). In case your team want to include sample collection and results of in-situ analysis performed by the rover it has to be performed again during the Scientific Exploration Sub-Task.

7.3.1.3 Deep Sampling Sub-Task

The main goal of this sub-task is to collect surface samples of the Martian soil that will be sent back to Earth for further analysis. Surface and Deep Sample Collection Sub-Tasks will be performed in the same time slot. The task scenario is as follows: 1. Arrive to the sampling location 2. Take a photo before sampling is started 3. Perform sampling (drilled-out core in case of deep sampling) 4. Place the collected sample/s in the onboard container 5. Measure weight of the collected sample/s 6. Take a photo after sampling is completed 7. Drive to the start/finish line

General requirements for the deep sampling are as follows: 1. The rover shall be equipped with a deep sample collection device that is able to pick up the surface sample from up to 30cm below the Mars Yard's surface and place it in the on-board container 2. The rover shall be equipped with a container/s allowing for the stable transport of the collected material over challenging terrain 3. The container/s shall keep the collected samples in a selected position and orientation and it shall prevent any movement 4. Deep sample must be collected by the rover. Both mass (>100g as a goal) and depth (>300mm as a goal) requirements are applicable to assure that scientifically valid samples and enough material are transported back to Earth. 5. The collected material must be further disposed in the onboard container and transported to the finish line (Martian Lander Module – MLMOD). 6. Weight of the sample must be measured by the onboard system of the rover. 7. Special tool or tools must be used to perform the tasks, e.g., drilling mechanism assembled on the robotic arm (any workarounds, e.g., drilling using rover's wheels) are prohibited. 8. Weight of the collected sample may be checked by judges at the finish line. Depth will be assessed based on the coloured layers of material in the drilling location. Moreover: * Max depth is not and will not be defined. Your task is to drill to at least 30cm. After that you can expect up to a few cm of soft sand/soil * There will be a coloured layer of soil at each scored level * Colour codes are provided for judges to assess how deep did you manage to drill and if you really drilled to that depth instead of drilling to the side

Foreseen colours of the layers for the Deep Sampling Sub-Task: * White / yellow sand > 30 cm * 25 cm < blue sand < 30 cm * 20 cm < purple sand < 25 cm * 15 cm < red sand < 20 cm * Soil with sand < 15 cm

DEEP SAMPLES COLLECTION PART – GENERAL COMMENTS: * Surface and Deep Sampling Sub-Tasks will be performed in a single run and sequence (one after another, sequence to be selected by teams) * Sampling zones will be predefined by judges. * Remember that in the collection part both surface and deep samples shall be collected! * Remember that in the deep sample collection part deep coloured sand samples shall be collected! * Remember about taking pictures before and after collecting samples * Measure weight of the collected sample/s. Depth from which the deep sample was collected has to be measured too * Science Surface and Deep Sample Collection Sub-Tasks are designed to solely check rover's ability to perform collection of various samples. These samples are not part of the Scientific Exploration activities (shall not be included in the report as the sampling location is defined by the organizer). In case your team want to include sample collection and results of in-situ analysis performed by the rover it has to be done during the Scientific Exploration Sub-Task. * In case of deep sampling drilling out (without placement in some kind of preliminary container but directly on the ground) and scooping the coloured layers with the robotic arm is forbidden. In this case you would 'contaminate' the deep sampling location for the other teams with the coloured soil (it will not be possible to scoop everything). In addition, if no coloured soil would be used (in a real life) you cannot be sure that you have managed to drill to the assumed depth - it would result in low technical excellence of your mechanism. You can drill out deep sample material, keep it in the tube around the drill and pour it in different location than deep sampling location (e.g. by simply changing rotation direction of the drill). Then you will not contaminate this location, and you can scoop the material. The perfect solution considers drilling out the sample and directly placing it into container that is used to perform weight measurement onboard and then by judges when the finish line has been reached.